A triangle incidence constraint at an exceptional compact hyperbolic Coxeter vertex

Coxeter Polytopes Frontier Project

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Abstract

At a finite vertex of type $E_7^3 \sqcup E_8$ in a compact hyperbolic Coxeter polytope in $\mathbb{H}^{29}$, we designate three two-face angles in each E_7 component. At least six of these nine faces are nontriangular. The proof reduces the actual third side of a triangle to a complete finite contact domain, checked by two exact implementations. All three angles force the same contact vector. The common finite vertex fixes the signs of its Lorentzian heights, and two such third sides in different components have a product strictly between consecutive native Coxeter bonds. No arithmeticity assumption is used. This is a local incidence theorem, not a new dimension bound or a face-charge saving.

1 Statement and geometric conventions

Wall normals have square 2 in a real Lorentz space of signature $(n, 1)$. A chamber is the intersection of their nonpositive halfspaces in the future hyperboloid. Products of distinct outward normals are nonpositive; a finite bond of label m has product $-2\cos(\pi/m)$. Parallel and ultraparallel walls have products -2 and less than -2, respectively.

We use the standard simplicity of compact Coxeter polytopes and the classification of finite Coxeter diagrams. In particular, a finite vertex of a compact n-polytope has exactly n independent incident walls. A whole E_8 has no connected finite extension, a connected finite one-root extension of E_7 is E_8, and a finite bond of label at least 6 is an isolated dihedral factor. These standard inputs are separated from the new geometric argument and finite computation below.

Number each E_7 by the simple edges

$$0 - 1 - 2 - 3 - 4 - 5, \qquad 2 - 6.$$

At a simple vertex, a pair of incident facets specifies the two-face whose supporting walls are all the other vertex facets.

Theorem 1. *Let v be a vertex of a compact hyperbolic Coxeter polytope in $\mathbb{H}^{29}$ with wall diagram $E_7 \sqcup E_7 \sqcup E_7 \sqcup E_8$. In each E_7 designate the pairs $(2,4), (2,5), (3,5)$. Among the three E_7 components, at most one supplies a triangular two-face at any of its designated pairs. Thus at most three of the nine designated faces are triangles, and at least six have four or more sides.*

2 The actual third side has no outside contacts

Let T be the positive span of the 29 roots at v. Suppose the two-face of a designated pair a, b is triangular, with third wall root c. Its other vertices have the old wall sets with a, respectively b,

replaced by c. These are actual positive-definite corners, not merely subdiagrams of a chosen local witness.

Lemma 2. *The root c is orthogonal to all old components except the marked E_7.*

Proof. The untouched E_8 appears whole at both new corners, so finite maximality forces orthogonality to it. If c contacts an untouched E_7 component F, then $F + c$ is a whole E_8 at each new corner. Finite maximality therefore forces c to be orthogonal to every other old root in those corners. Their union contains every old root. Hence the Gram matrix on all old roots and c is the positive-definite direct sum $(F + c) \sqcup (T - F)$, of rank 30. A Lorentz space of signature $(29, 1)$ has positive index 29, a contradiction. □

Lemma 3. *Every contact label of c with the marked E_7 belongs to $\{2, 3, 4, 5\}$.*

Proof. Every contact occurs at one of the two finite corners, so parallel and dotted products are excluded. A bond at least 6 would be an isolated $I_2(m)$ component. If its old endpoint j is white, it occurs in both corners. Its old neighbors would all have to lie in both omitted singletons $\{a\}$ and $\{b\}$, impossible in the connected E_7. If its old endpoint is a mark, that mark must be a leaf whose only neighbor is the other, omitted mark. The designated pairs are all nonadjacent, ruling this out. □

3 Complete exact triangle domains

Let G be the E_7 Gram matrix and $J = G^{-1}$. Write b for the seven products of c with that component. If e is the future unit vector at v, Lemma 2 gives

$$c = p + he, \qquad p \in T, \quad e^2 = -1, \quad h = -c \cdot e > 0, \qquad h^2 = bJb^T - 2.$$

The strict inequality holds because the third side does not pass through v. Thus a nonpositive computed h^2 cannot supply this side.

Lemma 4. *The wall set of a compact acute hyperbolic chamber contains no positive-semidefinite singular principal Coxeter diagram.*

Proof. Choose a connected singular component and positive kernel coefficients λ_i. Then $u = \sum_i \lambda_i r_i$ is null. A strict interior future timelike point x has $u \cdot x < 0$, so u is nonzero and future. Its products with the component vanish, and all its other wall products are nonpositive by acuteness. The resulting ideal point belongs to the closed chamber, contrary to compactness. This proof does not assume that the singular principal diagram was already known to support a face. □

Proposition 5. *For each designated pair, the unique possible triangle contact vector is*

$$c \cdot r_5 = -2\cos(\pi/5), \qquad c \cdot r_j = 0 \quad (j \neq 5).$$

Its height satisfies $h^2 = (1 + 3\sqrt{5})/4$.

Exact finite certificate. By Lemma 3, enumerate all 4^7 vectors for each pair. Test the two rank-seven finite corners obtained by omitting each mark. For every eligible vector compute $bJb^T - 2$. Discard nonpositive values; otherwise test the principal subsets of the eight-root diagram for the obstruction in Lemma 4. Exact results are:

Pair	Vectors	Finite eligible	$h^2 \le 0$	Singular PSD	Remaining
$(2,4)$	16,384	12	3	8	1
$(2,5)$	16,384	7	3	3	1
$(3,5)$	16,384	7	3	3	1

In all three rows the remaining vector is the one stated. The certificate contains every eligible vector and each explicit singular principal witness. The independent checker reconstructs the entire $49,152$-vector finite filter, recomputes the heights, verifies the singular witnesses, and exhausts all 255 nonempty principal subsets of each survivor.

All arithmetic takes place in $\mathbb{Q}(\sqrt{2},\sqrt{5})$ because the preceding geometric reduction forced labels at most 5. This is an exact computational representation, not an arithmeticity hypothesis. The producer's finite-diagram classifier and the checker's canonical rooted tree catalogue are separate implementations; their domain digests agree. Exact PSD tests use different Schur elimination orders. The old local witness catalogue also contains these three triangle recipes, but their sufficiency is not required for the necessary classification here. □

4 Coupling at the common vertex

Proof of Theorem 1. Suppose different E_7 components supply designated triangles with third roots $c_i = p_i + he$ and $c_j = p_j + he$. Their projections are orthogonal by Lemma 2. Both heights have the same positive sign, because the common vertex lies strictly on the interior side of both new walls. Their products are therefore

$$c_i \cdot c_j = -h^2, \qquad (c_i \cdot c_j)^2 = \frac{23+3\sqrt{5}}{8}.$$

The roots are distinct, having nonzero contacts on different old factors. Directed rational cosine enclosures give

$$\big(2\cos(\pi/11)\big)^2 < \frac{37}{10} < \frac{23+3\sqrt{5}}{8} < \frac{93}{25} < \big(2\cos(\pi/12)\big)^2.$$

The middle value is less than 4 and lies strictly between consecutive finite-bond squares. It is neither a finite Coxeter product nor a parallel or dotted product. Such two wall roots cannot coexist. At most one E_7 factor can therefore supply a designated triangle. There are three designated pairs per factor, and the nine pairs specify distinct faces. □

5 Reproduction, provenance, and limits

The complete exact certificate is `results/fresh_compact29_three_triangle_star_constraint_20261018.json`. Its producer, report, inherited data and full independent checker are listed in `docs/compact29_triangle_constraints_map.md`. Reproduction uses the producer's nonmutating `-verify` mode and the standalone saved checker. The checker uses only the Python standard library and imports no project module. It checks the finite certificate, not the entire theory of hyperbolic polytopes from axioms.

The theorem is internally checked by exact computation and written mathematical review. External refereeing, priority and full proof-assistant verification are separate. No new compact dimension bound follows at this stage: a quadrilateral can still consume the current face-charge allowance. Improving the bound requires a further charging or incidence theorem covering the remaining faces and all relevant vertex types. No finite-volume construction is claimed.